

ORIGINAL ARTICLE

Niche Differentiation Among Large Pelagic Fishes in the Southwest Atlantic Revealed by Stable Isotope Analysis

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ABSTRACT

Understanding the trophic ecology of large pelagic fishes is essential to assess their ecological role in marine ecosystems and to develop effective conservation strategies. In the present study, we aimed to provide new information on the trophic ecology of eight large pelagic fish species (*Thunnus atlanticus* (blackfin tuna), *Thunnus albacares* (yellowfin tuna), *Katsuwonus pelamis* (skipjack tuna), *Sarda sarda* (Atlantic bonito), *Euthynnus alletteratus* (little tunny), *Coryphaena hippurus* (dolphinfish), *Istiophorus albicans* (Atlantic sailfish), and *Kajikia albida* (white marlin)) using carbon and nitrogen stable isotope analysis. The isotopic data were used to determine the foraging range and trophic position of the species and to assess the isotopic niche widths and resource partitioning between them. Pelagic fish samples ($N=163$) were acquired from the Cabo Frio commercial fleet, Rio de Janeiro, Brazil, between February 2009 and January 2010. We found significant differences in both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values among the species. *S. sarda* appears to feed at higher trophic levels and shows isotopic niche segregation with the species analyzed. However, there was a high degree of niche overlap between the other species, with the exception of the genus *Thunnus*. The isotopic niche width varied among species, with *C. hippurus* and *E. alletteratus* showing the widest niches, while *S. sarda* had the narrowest. These results highlight relevant aspects of the trophic dynamics of species of high commercial value in a region that has not been previously studied. The information contained here can serve as a basis for future conservation and management strategies for these species in the southwest Atlantic.

1 | Introduction

Large pelagic fish species exhibit wide-ranging distributions in the tropical and subtropical oceans that support extensive commercial and recreational fisheries worldwide (Block and Stevens 2001;

Laiz-Carrión et al. 2019). Moreover, these organisms are apex predators and play a fundamental role in marine ecosystems through ecological interactions (Heithaus et al. 2008; Bornatowski et al. 2017). Large pelagic fishes exert top-down effects on marine communities, and thus, any population decline can have

large impacts on ecosystem structure and functioning (Baum and Worm 2009; Navarro et al. 2020; Pintado-Herrera et al. 2024).

Understanding the feeding ecology of marine organisms is essential to determine their ecological role in ecosystems and to provide information on their interactions regarding the partitioning of resources, the flow of energy and contaminants, and the effects of competition and predation on food web structures (Bornatowski et al. 2017; Albuquerque et al. 2019; Navarro et al. 2020). In this sense, stable isotope analysis (SIA) has been extensively applied to study the diets and trophic relationships of species in marine food webs (Peterson and Fry 1987). Stable nitrogen isotopes (expressed as $\delta^{15}\text{N}$) exhibit significant enrichment through trophic transfers and thus can be used as a proxy for estimating the trophic position of organisms (Layman et al. 2007). Conversely, stable carbon isotopes ($\delta^{13}\text{C}$) vary substantially among primary producers with different photosynthetic pathways (e.g., C3 vs. C4 plants) but change little with trophic transfers (DeNiro and Epstein 1978). Accordingly, $\delta^{13}\text{C}$ can reflect the primary source of carbon in the food web, allowing for differentiation between organisms associated with either benthic or pelagic systems and coastal or oceanic environments (Peterson and Fry 1987).

The marine fish species investigated in this study are widely distributed in tropical and subtropical regions of the world, occupying mainly epipelagic regions, with some species making sporadic dives into the mesopelagic zone (Holland et al. 1990; Madigan et al. 2021). These fish species are mostly generalists, feeding on fish, crustaceans, and cephalopods. This is true for the species in the family Scombridae examined here, including *Thunnus atlanticus* (blackfin tuna), *Thunnus albacares* (yellowfin tuna), *Katsuwonus pelamis* (skipjack tuna), *Euthynnus alletteratus* (little tunny), and *Sarda sarda* (Atlantic bonito), whose diet has been well documented in different regions. Based on previous studies, the species in the genus *Thunnus* and the dolphinfish (*Coryphaena hippurus*, Coryphaenidae) reported here have a diet composed of teleost fish, mainly species in the families Exocoetidae, Myctophidae, and Bramidae (Headley et al. 2009; Vaske and Lessa 2004; Albuquerque et al. 2019), as well as crustaceans and cephalopods (Malone et al. 2011; Gorni et al. 2013; Varela et al. 2017). The other species in the family Scombridae studied here show differences in prey items as evidenced by previous studies of stomach contents. *K. pelamis* has a high consumption of lanternfish and euphausiid crustaceans in different regions (Vilela 1990; Matsuura and Andrade 2000; Coletto et al. 2021), whereas *E. alletteratus* and *S. sarda* feed on species in the genus *Sardinella* as well as species in the family Engraulidae, such as anchovies (Campo et al. 2006; Fletcher et al. 2013; González-Rojas et al. 2024). Finally, the species of the family Istiophoridae studied here have a diet composed of teleost fish, mainly in the families Bramidae, Clupeidae, Dactylopteridae, Exocoetidae, and Scombridae, as well as cephalopods such as the squid *Ornithoteuthis antillarum* (Nakamura 1985; Satoh et al. 2004; Junior et al. 2004; Bahou et al. 2018).

Although several studies have focused on the diet and trophic relationships of large pelagic fishes (Varghese et al. 2014; Navarro et al. 2017; Navarro et al. 2020; Albuquerque et al. 2019; Rosas-Luis et al. 2021), the trophic role that these species play in the southwestern Atlantic region—where research is still scarce—remains unclear (Gorni et al. 2013, 2017). In the present study, we aimed to investigate the trophic ecology of eight large pelagic

fish species, namely, *T. atlanticus* (blackfin tuna), *T. albacares* (yellowfin tuna), *S. sarda* (Atlantic bonito), *E. alletteratus* (little tunny), *K. pelamis* (skipjack tuna), *C. hippurus* (dolphinfish), *Kajikia albida* (white marlin, Istiophoridae), and *Istiophorus albicans* (sailfish, Istiophoridae) in the Southwest Atlantic (SWA) Ocean off the coast of the state of Rio de Janeiro, Brazil. We focused our investigation on understanding aspects of the isotopic niche width of these sympatric species by explaining the partitioning of resources between them using carbon and nitrogen SIA. Furthermore, we investigated potential shifts in foraging areas and trophic levels associated with increases in body size. Our results offer new insights into the trophic ecology of tuna and tuna-like species in the SWA Ocean, providing critical information for the sustainable management of these species.

2 | Material and Methods

2.1 | Sampling

The pelagic fish specimens ($N=163$) were acquired between February 2009 and January 2010 from Cabo Frio's commercial fleet, which operates in the southwestern Atlantic Ocean in Brazil. Small boats and different fishing gears were used for sampling, including surface longline, mid-water longline, drop line, hand line, and purse seine in locations with depths ranging from 17 to 2000 m and distances from the shore of 1–190 km (Figure 1).

Most of the fish were caught whole, but some specimens could only be acquired from parts of the fish, resulting in a lack of biological data for these individuals. The target species were the blackfin tuna (*T. atlanticus*), yellowfin tuna (*T. albacares*), skipjack tuna (*K. pelamis*), Atlantic bonito (*S. sarda*), little tunny (*E. alletteratus*), dolphinfish (*C. hippurus*), Atlantic sailfish (*I. albicans*), and white marlin (*K. albida*). Each fish specimen was weighed, measured, and dissected. Next, an aliquot of muscle was stored and frozen at -20°C until analysis.

2.2 | SIA

A portion of the dorsal muscle was sampled from each individual. The aliquots were freeze-dried and macerated until a homogeneous powder was formed. Approximately 0.25 mg of each dried sample was weighed in a tin capsule. The carbon and nitrogen isotope ratios were determined using a DELTA V isotope ratio mass spectrometer (IRMS; Thermo Fisher Scientific, Waltham, MA, USA) coupled to a Flash EA C–N–S elemental analyzer (Thermo Fisher Scientific). The isotope ratio is expressed using the delta notation in parts per thousand (‰), according to the formula below:

$$\delta X = \left[\left(R_{\text{sample}} / R_{\text{standard}} \right) - 1 \right] \times 1000, \quad (1)$$

where X is either $\delta^{13}\text{C}$ or $\delta^{15}\text{N}$, R is given by the $^{13}\text{C}/^{12}\text{C}$ or $^{15}\text{N}/^{14}\text{N}$ ratio, and the standard reference values are VPDB (Vienna Pee Dee Belemnite) for carbon and atmospheric air for nitrogen. Certified reference materials (USGS24, Caffeine, and IAEA-N-2), internal replicates, and analytical blanks were used to ensure the accuracy and precision of the results. The analytical precision was $<0.2\text{‰}$ for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$.

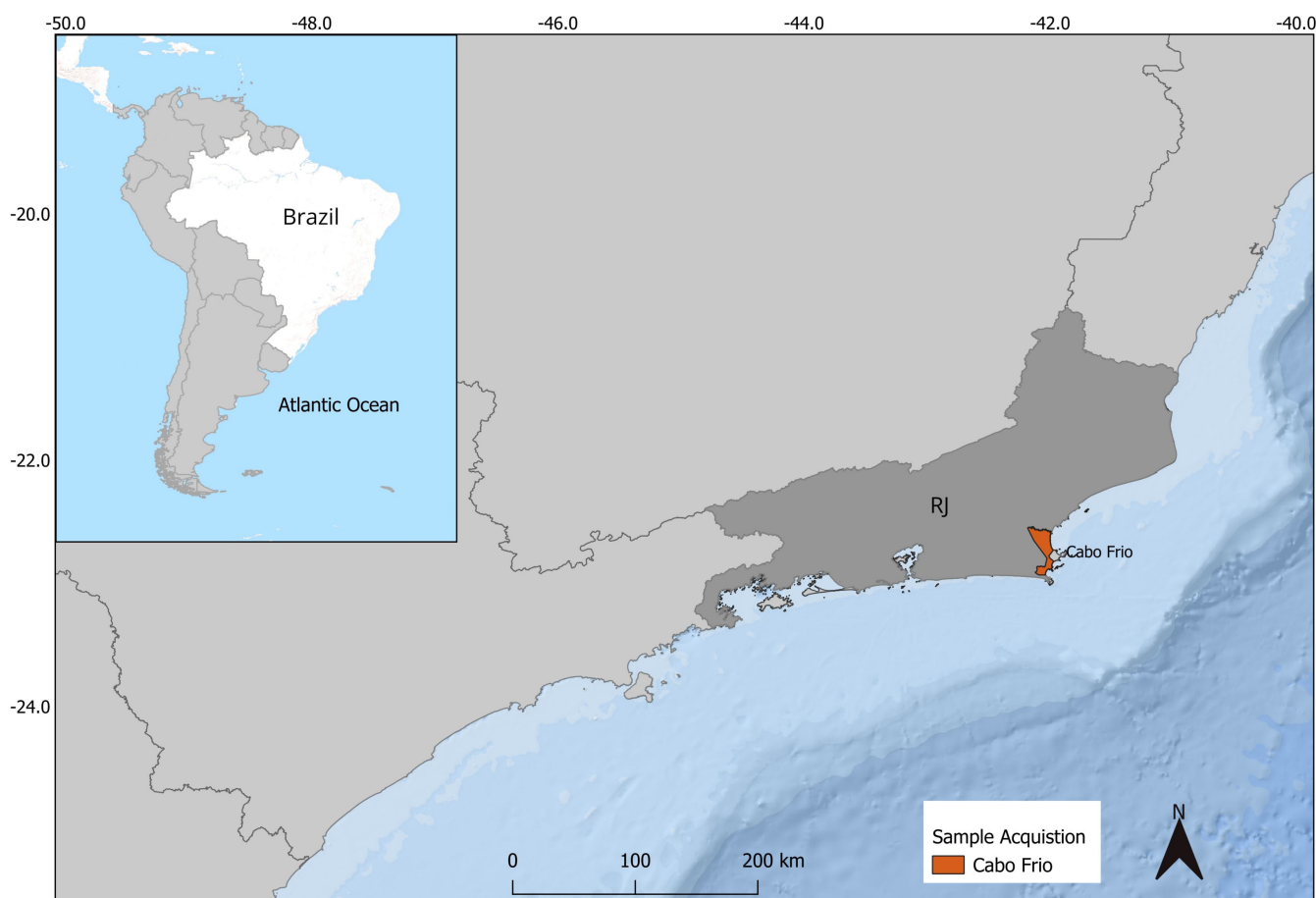


FIGURE 1 | Map showing the Southwest Atlantic Ocean where the pelagic fish samples were collected. The region in red indicates where the specimens were purchased.

2.2.1 | Lipid Correction

The lipid fraction in the tissues can have depleted ^{13}C values (DeNiro and Epstein 1977), which can affect the ^{13}C values observed in the animal tissues. For this reason, for individuals with a C:N ratio > 3.5 , chemical lipid extraction was performed as described by Bligh and Dyer (1959) and modified and described by Logan et al. (2008). Approximately 0.1 g of muscle tissue was weighed into a 10-ml glass vial. In each tube, 2 mL of the chloroform: methanol ratio (2:1, v:v) was added. Next, the samples were homogenized for 30 s in a vortex. After 30 min, the supernatant (containing the lipid and the solvent mixture) was removed. This procedure was repeated for at least three washes until the supernatant was colorless. Finally, the samples were dried in an oven at 60°C for 24 h. After drying, the samples were weighed and reanalyzed to measure the corrected $\delta^{13}\text{C}$ values ($n = 10$). The values of the C:N ratio before lipid extraction varied between 3.54 and 4.44. After lipid extraction, the C:N ratio varied between 3.15 and 3.42.

For five samples (one from each of the following species: *T. albacares*, *S. sarda*, and *K. albida*; and two for *I. albicans*), where it was not possible to obtain the weight required for chemical extraction, a mathematical correction was made according to the equation by Post et al. (2007):

$$\delta^{13}\text{C}_{\text{normalized}} = \delta^{13}\text{C}_{\text{untreated}} - 3.32 + 0.99 * \text{C:N}. \quad (2)$$

2.3 | Statistical Analysis

The data were tested for normality and homogeneity using the Shapiro–Wilk and Levene's tests, respectively. When data were normally distributed, one-way analysis of variance (ANOVA) was used to verify the differences in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ between the species. When significant differences were observed, Tukey's test was applied to determine which groups differed from each other. Differences were considered significant when the p value was < 0.05 . The Stable Isotope Bayesian Ellipses in R (SIBER) program (Jackson et al. 2011) was used to assess the isotopic niche of each species. The total area occupied by the polygon (TA), the standard area of the ellipses (SEA), and the standard area of the ellipses corrected for the low number of samples (SEAc) were calculated, as well as the degree of niche overlap between the species. The degree of overlap and its associated metrics (Layman et al. 2007) were also analyzed using the SIBER package, where CD represents the average distance to the centroid, MMND the average distance to the nearest neighbor, and SDNND the standard deviation of the MNND metric.

The dissimilarities in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values allowed the species to be grouped using hierarchical clustering, which was performed by calculating the Euclidean distances using Ward's method. For those species where there was sufficient information on standard length (SL), Spearman's correlation was used to assess

the correlation between the isotopic values ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) and the SL. The R statistical software version 4.1.1 for Windows (R Core Team 2020) was used for all data processing.

3 | Results

A total of 163 specimens from eight large pelagic fish species were analyzed (Table 1). Results for SIA revealed that mean values for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ in muscle tissue ranged from -19.3‰ to -16.0‰ and 9.2‰ to 16.9‰ , respectively (Table 1). Significant differences were found in mean $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values among species (ANOVA, $p < 0.05$). Mean $\delta^{13}\text{C}$ was significantly higher in *T. albacares* relative to the other species (Tukey's test, $p < 0.05$; Table 2), except *T. atlanticus*. Conversely, mean $\delta^{13}\text{C}$ was significantly lower in *C. hippurus* than in *S. sarda*, *K. pelamis*, *T. atlanticus*, and *T. albacares* (Tukey's test, $p < 0.05$; Table 2).

Additionally, mean $\delta^{15}\text{N}$ was highest in *S. sarda* (Tukey's test, $p < 0.05$; Table 2) and was significantly lower in *T. albacares* relative to *T. atlanticus*, *E. alletteratus*, *S. sarda*, *C. hippurus*, and *I. albicans* (Tukey's test, $p < 0.05$; Table 2).

Hierarchical cluster analysis distinguished two main groups of large pelagic fishes based on their isotopic values. One group consisted of species from the family Scombridae, except *S. sarda*, and was further subdivided into two subgroups. The other group included species from the families Istiophoridae and Coryphaenidae and *S. sarda* (Figure 2). Within the Scombridae group, one subgroup contained only species from the genus *Thunnus*, whereas the other included *E. alletteratus* and *K. pelamis*. Similarly, the second main group was divided into two subgroups: one subgroup composed of species from the family Istiophoridae and *C. hippurus* and the second subgroup containing *S. sarda* (Figure 2).

TABLE 1 | Mean and standard deviation (SD) values for standard length (SL, cm) and mean $\pm$ SD, median, minimum (Min), and maximum (Max) values for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ in muscle tissue of large pelagic fishes in the Southwest Atlantic Ocean, Brazil.

Species	<i>n</i>	Standard length (cm) Mean $\pm$ SD	$\delta^{13}\text{C}$ (‰) Mean $\pm$ SD Min/Max	$\delta^{15}\text{N}$ (‰) Mean $\pm$ SD Min/Max
Scombridae				
<i>Thunnus atlanticus</i>	27	48.6 $\pm$ 15.3	-17.4 ± 0.5 $-18.9/-16.4$	9.6 $\pm$ 1.2 7.0/11.8
<i>Thunnus albacares</i>	25	49 $\pm$ 15.8	-17.0 ± 0.4 $-18.2/-16.0$	9.2 $\pm$ 1.2 7.2/12.0
<i>Katsuwonus pelamis</i>	34	46.8 $\pm$ 10.4	-18.0 ± 0.4 $-18.7/-16.9$	10.5 $\pm$ 1.8 6.5/14.1
<i>Euthynnus alletteratus</i>	9	45.3 $\pm$ 13.0	-18.1 ± 0.4 $-18.6/-17.1$	11.7 $\pm$ 1.9 7.3/13.2
<i>Sarda sarda</i>	19	38.5 $\pm$ 5.3	-17.5 ± 0.4 $-19.1/-17.1$	14.5 $\pm$ 1.7 9.6/16.9
Coryphaenidae				
<i>Coryphaena hippurus</i>	29	66.1 $\pm$ 31.9	-18.4 ± 0.6 $-19.2/-16.8$	10.8 $\pm$ 1.4 7.3/13.8
Istiophoridae				
<i>Kajikia albidia</i>	6	114 $\pm$ 48.0	-18.0 ± 0.5 $-18.5/-17.2$	11.0 $\pm$ 1.7 10.1/12.4
<i>Istiophorus albicans</i>	14	72.3 $\pm$ 17.21	-17.8 ± 0.5 $-19.3/-17.1$	10.9 $\pm$ 1.3 8.6/13.9

Note: Total length (TL) was used for *C. hippurus* and *I. albicans* because there was no information on standard length (SL) for these species.

TABLE 2 | Results of the Tukey post hoc test for multiple comparisons of $\delta^{13}\text{C}$ (upper right) and $\delta^{15}\text{N}$ (lower left). Values in bold indicate $p < 0.05$.

	TAT	TALB	KP	SS	EA	CH	IA	KA
TAT		0.01	0.01	0.91	0.06	0.00	0.58	0.12
TALB	0.99	0.00	0.00	0.00	0.00	0.00	0.00	0.00
KP	0.76	0.23		0.58	0.99	0.02	0.99	1.00
SS	0.00	0.00	0.00		0.55	0.00	0.97	0.81
EA	0.10	0.02	0.63	0.00		0.73	0.99	0.99
CH	0.06	0.00	0.79	0.00	0.99		0.45	0.14
IA	0.04	0.00	0.50	0.00	1.00	0.99		1.00
KA	0.77	0.44	0.99	0.00	0.99	0.99	0.99	

Abbreviations: CH, *C. hippurus*; EA, *E. alletteratus*; IA, *I. albicans*; KA, *K. albidia*; KP, *K. pelamis*; SS, *S. sarda*; TALB, *T. albacares*; TAT, *T. atlanticus*.

3.1 | Niche Breadth and Niche Overlap

Niche breadth, as measured by the SEAc, was highest in *C. hippurus* (SEAc = 3.0‰²), followed by *E. alletteratus* (SEAc = 2.6‰²), and was lowest in *S. sarda* (SEAc = 1.4‰²) (Figure 3, Table 3). *K. pelamis* and *K. albida* had the highest and lowest $\delta^{15}\text{N}$ range, respectively (Table 2). The highest and lowest CR values, respectively, were recorded for *T. atlanticus* (2.4) and *K. albida* (1.3), and CD values were similar among species (Table 3). The lowest CD value was associated with *K. albida* (0.8), and the other species had CD values ranging from 1.1 to 1.5 (Table 3).

We found high niche overlap in feeding habits among some species. *E. alletteratus* showed the highest probability of niche overlap with *K. pelamis* (86.8%), *C. hippurus* (86.8%), *I. albicans* (78.2%), *T. atlanticus* (65.8%), and *K. albida* (63.8%) (Table 4, Figure 3). Conversely, low niche overlap was observed between *S. sarda* and the other species (<40%). In addition, *T. albacares*

also showed niche segregation with the other species, except for *T. atlanticus*, for which it had an 84.3% probability of being present in the isotopic niche of the latter (Table 4).

A positive correlation between standard length and $\delta^{15}\text{N}$ values was found for *T. atlanticus*, *K. pelamis*, *E. alletteratus*, and *S. sarda* (Spearman's correlation, $p < 0.05$; Figure 4). Additionally, there was a positive correlation between standard length (or total length) and $\delta^{13}\text{C}$ for *T. albacares* and *C. hippurus* (Spearman's correlation, $p < 0.05$; Figure 5). For the other species, no significant correlations were found for any of the isotopic values (Spearman's correlation, $p > 0.05$).

4 | Discussion

This study revealed variations in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values among eight sympatric species of large pelagic fish in the southwestern

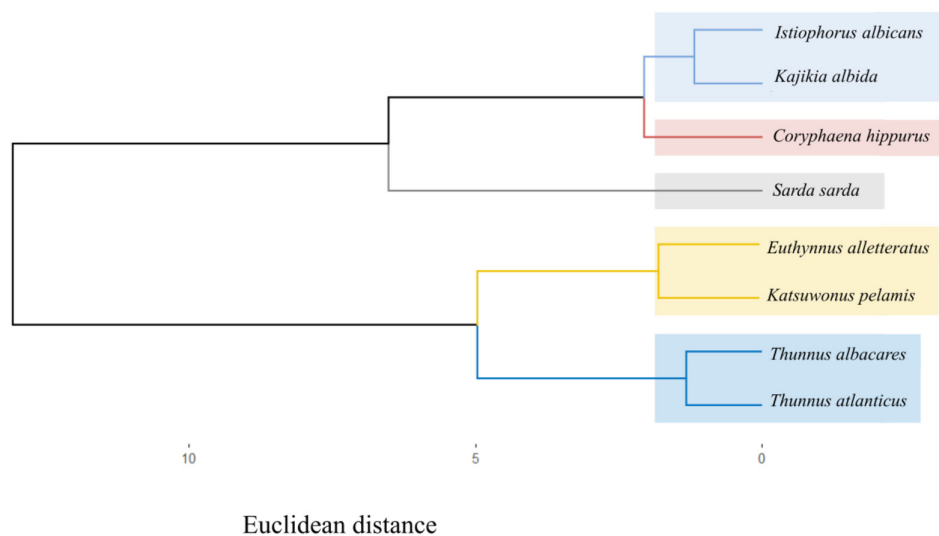


FIGURE 2 | Dendrogram for hierarchical cluster analysis using Ward's method based on $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values from the muscle of pelagic fish species from the Southwest Atlantic Ocean, Brazil.

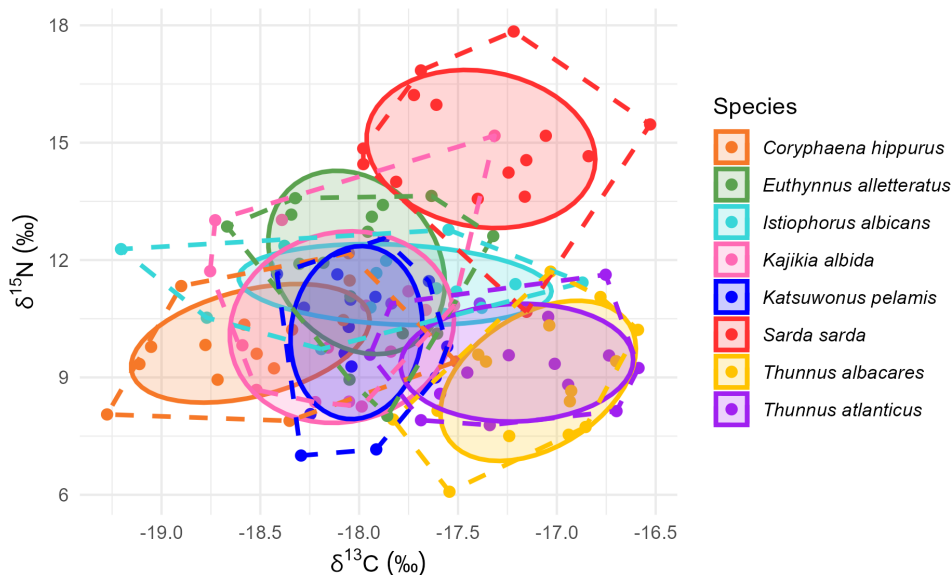


FIGURE 3 | Standard ellipse area corrected for sample size (SEAc) for pelagic fish species from the Southwest Atlantic Ocean, Brazil.

TABLE 3 | Isotopic niche areas and population metrics for pelagic fishes from the Southwest Atlantic Ocean, Brazil.

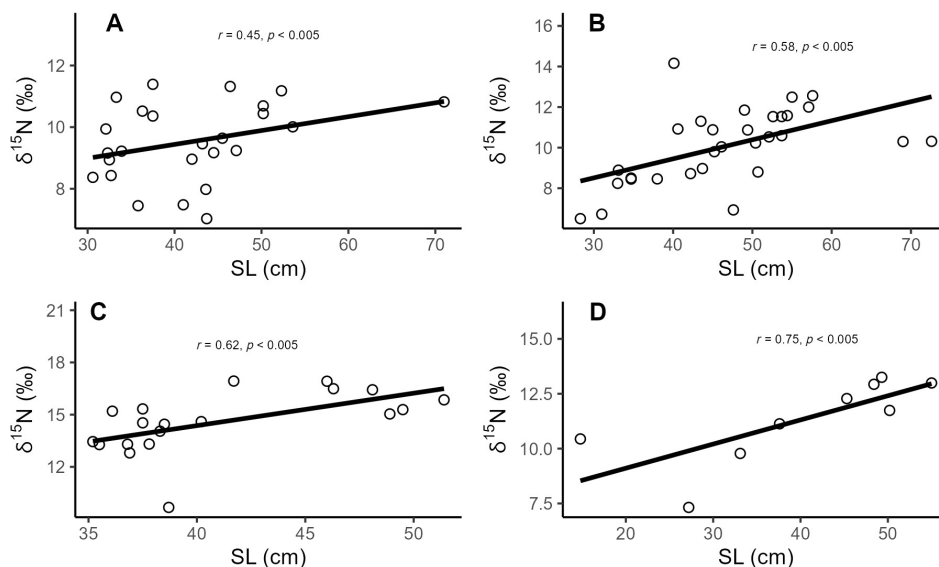
Species	TA	SEA	SEAc	NR	CR	CD	MNND	SDNND
<i>Thunnus atlanticus</i>	6.8	1.9	2.1	4.8	2.4	1.1	0.4	0.2
<i>Thunnus albacares</i>	4.7	1.6	1.7	4.7	1.7	1.2	0.3	0.3
<i>Katsuwonus pelamis</i>	9.1	2.4	2.5	7.6	1.7	1.5	0.2	0.3
<i>Euthynnus alletteratus</i>	4.6	2.3	2.6	5.9	1.4	1.5	0.8	0.7
<i>Sarda sarda</i>	4.0	1.4	1.5	7.2	1.9	1.3	0.4	0.6
<i>Coryphaena hippurus</i>	10.0	2.9	3.0	6.5	2.3	1.2	0.2	0.2
<i>Kajikia albida</i>	2.3	1.8	2.5	2.2	1.3	0.8	0.8	0.1
<i>Istiophorus albicans</i>	5.1	2.2	2.5	3.4	2.2	1.1	0.5	0.3

Abbreviations: CD: mean distance to centroid; CR: $\delta^{13}\text{C}$ range; NND: mean nearest neighbor distance; NR: $\delta^{15}\text{N}$ range; SDNND: standard deviation of nearest neighbor distance; SEA: standard ellipse area (‰^2); SEAb: Bayesian standard ellipse area (‰^2); SEAc: standard ellipse area corrected for sample size (‰^2); TA: total area (‰^2) of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values.

TABLE 4 | Posterior mean ($\alpha=0.95$) for isotopic niche overlap (%) between each pair of pelagic fish species from the Southwest Atlantic Ocean, Brazil. Probabilities > 60% are highlighted.

Species	TAT	TALB	KP	SS	EA	CH	KA	IA
<i>Thunnus atlanticus</i>	—	70.4	81.5	6.6	68.4	75.1	50.2	66.1
<i>Thunnus albacares</i>	84.3	—	46.8	1.0	34.3	42.3	33.1	53.2
<i>Katsuwonus pelamis</i>	76.9	32.5	—	23.6	80.2	87.2	57.3	71.6
<i>Sarda sarda</i>	11.4	1.5	40.1	—	37.2	35.2	22.5	37.0
<i>Euthynnus alletteratus</i>	65.7	25.7	86.8	36.9	—	86.8	63.7	78.2
<i>Coryphaena hippurus</i>	54.9	17.2	74.9	26.5	66.3	—	55.8	81.6
<i>Kajikia albida</i>	20.8	3.8	33.0	4.8	79.3	90.5	—	94.1
<i>Istiophorus albicans</i>	50.2	32.0	59.0	23.5	53.10	72.1	56.7	—

Abbreviations: CH: *C. hippurus*; EA: *E. alletteratus*; IA: *I. albicans*; KA: *K. albida*; KP: *K. pelamis*; SS: *S. sarda*; TALB: *T. albacares*; TAT: *T. atlanticus*.

**FIGURE 4** | Correlation between standard length (SL) and $\delta^{15}\text{N}$ values for pelagic fish species *Thunnus atlanticus* (A), *Katsuwonus pelamis* (B), *Sarda sarda* (C), and *Euthynnus alletteratus* (D) from the Southwest Atlantic Ocean, Brazil.

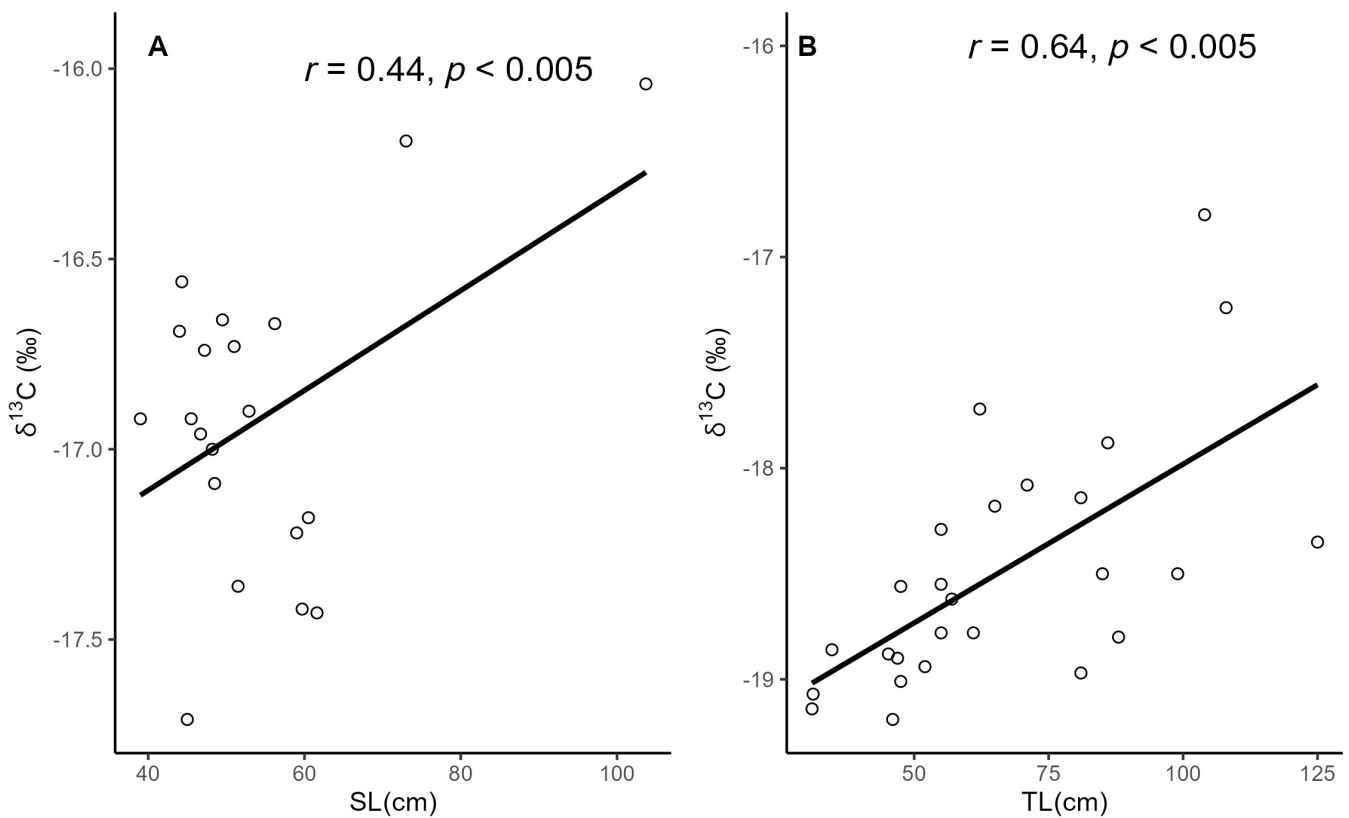


FIGURE 5 | Correlation between standard length (SL) and $\delta^{13}\text{C}$ values for pelagic fish species *Thunnus albacares* (A) and *Coryphaena hippurus* (B) from the Southwest Atlantic Ocean, Brazil.

Atlantic Ocean off the coast of Brazil. These variations reflect differences in their trophic ecology and resource partitioning.

Differences in $\delta^{13}\text{C}$ values reflect the differences in the primary sources of carbon in trophic webs, where coastal systems have values higher than the pelagic systems (Peterson and Fry 1987). The average $\delta^{13}\text{C}$ values showed interspecific differences, suggesting that the species in the study explore different foraging areas. The higher average $\delta^{13}\text{C}$ values observed for *T. albacares* and *T. atlanticus*, compared to the others herein investigated, are associated with the coastal habit of these species (Dragovich and Potthoff 1972; Bard et al. 2002; Fenton et al. 2015). However, subadult and adult individuals are commonly observed in oceanic regions (Costa et al. 2005; Domingo et al. 2008). Tagging studies also report that both species make sporadic deep dives in the mesopelagic region when they reach larger sizes (Weng et al. 2009; Fenton et al. 2015). In the present study, it is important to note that the data set of *T. albacares* and *T. atlanticus* primarily consisted of juveniles, which may have skewed the $\delta^{13}\text{C}$ values in these species towards higher values.

Although differences were observed in the $\delta^{13}\text{C}$ values between Scombridae species, cluster analysis revealed the formation of a group composed only of species in the family Scombridae. This result suggests that these species may occupy similar trophic levels and share similar carbon sources at some stage of their life cycle. Moreover, this finding is supported by studies of the behavior of these species, which reported shoaling between *T. atlanticus*, *T. albacares*, *E. alletteratus*, and *K. pelamis*, indicating

that they may compete for prey among themselves in the same area (Collette and Nauen 1983; Fenton et al. 2015). Thus, the separation of the species in the genus *Thunnus* in this study, evidenced by the higher average $\delta^{13}\text{C}$ values, may be associated with the predominance of juveniles in the sample, which have a more coastal habit in their juvenile phase, as discussed above.

Based on the similarities in mean $\delta^{13}\text{C}$ values, the species of the families Istiophoridae and Coryphaenidae were grouped together in the cluster analysis. The low $\delta^{13}\text{C}$ values found for *C. hippurus*, *I. albicans*, and *K. albidus* suggest that these species use oceanic waters as feeding habitats (Oxenford and Hunte 1999; Moltó et al. 2020). Also, the similarity in average $\delta^{13}\text{C}$ values between these species may be related to the consumption of prey from similar carbon sources, which is consistent with previous studies in the North Atlantic, where *I. albicans* and *K. albidus* exhibited high overlap in their prey items (Satoh et al. 2004). In the same study, *C. hippurus* also showed similarities in the prey consumed by the two species. Madigan et al. (2021) grouped *C. hippurus* and *I. albicans* in the same clade based on depth distribution, indicating that these species remain, for most of the time, in waters above the thermocline in the South Atlantic. Future studies using stomach content analysis and biological markers could help clarify the use of areas by these species and their influence on diet.

Finally, the formation of a distinct group for *S. sarda* was mainly driven by its higher $\delta^{15}\text{N}$ value compared to other species, showing that this species feeds at higher trophic levels.

Studies in other regions have also reported that *S. sarda* occupies a higher trophic level compared to other pelagic fish species in the Mediterranean Sea (Navarro et al. 2017; Muñoz-Lechuga et al. 2024; Pintado-Herrera et al. 2024). This result may be associated with a specialist strategy adopted by *S. sarda*, which feeds mainly on clupeid fish, gadids, and anchovies (ICCAT 2021) but may also include tuna species such as *E. alletteratus* and the bullet tuna (*Auxis rochei*) in its diet, as well as individuals of its own species (Zusser 1954; Muñoz-Lechuga et al. 2024). This behavior may explain its higher trophic position in relation to the other species in this study, but to the best of our knowledge, no studies have investigated its diet in the South Atlantic. Nevertheless, a study in the same region reported higher mercury concentrations in *S. sarda* when compared to other tuna species, which was attributed to its piscivorous feeding habit, with the consumption of high-trophic-level species (Manhães et al. 2020).

Similarly, the low mean $\delta^{15}\text{N}$ values found for *T. albacares* and *T. atlanticus*—the lowest of all species examined in this study—may have been affected by the higher proportion of juveniles in the sample or by the fact that, in the Southwest Atlantic, *T. atlanticus* preferentially forages for crustaceans and cephalopods as a way of avoiding competition with other sympatric pelagic species, as observed in the São Pedro and São Paulo Archipelago (Albuquerque et al. 2019). However, studies from other regions have shown that *T. albacares* and *T. atlanticus* preferentially feed on fish, especially mesopelagic and benthopelagic species that migrate to the surface, such as fish from the families Myctophidae and Trichiuridae (Headley et al. 2009). Future studies may shed light on the dietary patterns of *T. atlanticus* and *T. albacares* in the Southwest Atlantic Ocean by combining stomach content and SIA.

4.1 | Niche Width and Overlap

The broad niches observed for *C. hippurus*, *E. alletteratus*, *K. pelamis*, *K. albida*, and *I. albicans* were likely due to the consumption of a wide variety of prey with different carbon sources and/or that are at different trophic levels (Bearhop et al. 2004). The high TA and NR values for *C. hippurus* support the trophic diversity related to its generalist feeding habit and high consumption of prey, which can include fish from the families Exocoetidae, Hemiramphidae, and Balistidae, as well as cephalopods and crustaceans (Palko et al. 1982; Rudershausen et al. 2010; Varghese et al. 2014; Albuquerque et al. 2019). The same pattern was also observed for *K. pelamis*, which had high TA and NR values, indicating a high diversity in prey consumption. The large variation in the range of nitrogen values observed for this species may be associated with the consumption of organisms from different trophic levels in the South Atlantic Ocean, including euphausiid crustaceans and fish from the family Myctophidae (Zavala-Camim 1982; Matsuura and Andrade 2000; Coletto et al. 2021; Monteiro 2023).

Narrower isotopic niches, however, may reflect a pattern of consumption of specific prey (Lowry et al. 1991). This strategy, commonly referred to as “opportunistic specialist”, can be attributed to *S. sarda*, which had the lowest value of isotopic niche breadth among the species studied. This result, together with its high

$\delta^{15}\text{N}$ values, reinforces the specialist strategy of this species, as has been widely documented in the literature.

K. albida had a 94.1% and 90.5% chance of overlapping with *I. albicans* and *C. hippurus*, respectively, indicating that these species are possibly sharing resources (Bearhop et al. 2004). The data available in the literature on these species is still scarce, and no studies have been identified that use stable isotopes to assess resource partitioning between the three species together. However, studies based on diet characterization have reported that *K. albida* and *I. albicans* feed on the same prey items in the Northeast region of Brazil, with fish from the families Bramidae and Scombridae comprising the highest proportion of prey fish in the diet of these two species (Satoh et al. 2004). Also in that study, *C. hippurus* was found to have a highly psychoactive diet of species from the families Exocoetidae and Balistidae. In a study by Logan et al. (2013) in the North Atlantic, *C. hippurus* was found to be at a lower trophic level than *I. albicans*, which was attributed to *C. hippurus*' consumption of prey associated with *Sargassum* spp., a lower trophic level organism.

C. hippurus also showed a probability of isotopic niche overlap with *K. pelamis* (74.9). To date, only one study has reported the degree of overlap between these two species, with high levels of overlap associated with the consumption of similar prey, mainly small pelagic fish (Hildebrand 2022). High probabilities of overlap in the isotopic niche have also been reported between *E. alletteratus* and *K. pelamis*. A study in Florida observed that the two species had similar diets, with a high contribution of fish from the family Clupeidae in the diet of both species in that region (Moore 2014). Moreover, these species often form shoals to forage, increasing the likelihood that both species to consume the same prey or prey of similar isotopic composition (Moore 2014; Fenton et al. 2015).

Finally, a high probability of overlap was also found between *T. atlanticus* and *K. pelamis* (81.5%), indicating that, in the Southwest Atlantic, the two species share resources between them. This is the first study to date to assess the degree of overlap between these two species. The available data on their diet show that both species consume fish from the family Carangidae in different regions (Dragovich 1971; Manooch and Mason 1983; Ankenbrandt 1985). It is important to consider, however, that these studies have been conducted in different regions and that the high degree of overlap between *T. atlanticus* and *K. pelamis* may possibly be more related to the consumption of prey with similar isotopic compositions, in the hypothesis that the two species share foraging areas (Fenton et al. 2015).

In contrast, *S. sarda* showed isotopic niche segregation with the other analyzed species, indicating food resource partitioning. This pattern is related to the differentiation in average $\delta^{15}\text{N}$ values, suggesting that the species consumes prey from higher trophic levels. The segregation of *S. sarda* in relation to other pelagic fish has been widely documented in other studies and is attributed to the consumption of prey from higher trophic levels and the greater specialization of its diet, which is predominantly piscivorous. This specialization contrasts with the generalist strategy often reported for tunas and related species (Varela et al. 2019; Navarro et al. 2020; Muñoz-Lechuga et al. 2024; Pintado-Herrera et al. 2024).

4.2 | Comparison Between Standard Length and Isotopic Values

Changes in the diet of scombrids throughout ontogenetic development are well documented in the literature and may be a strategy adopted to reduce intraspecific competition (Graham et al. 2007; Weng et al. 2016; Fuller et al. 2021). Changes in the diet of tunas and tuna-like species, however, are more heavily determined by the size of the individual, which does not necessarily reflect the size reached at sexual maturity, and there can be significant changes in diet between size ranges at the same stage of the life cycle (Graham et al. 2007). In this study, significantly positive correlations were observed between $\delta^{15}\text{N}$ and standard length for the species *T. atlanticus*, *K. pelamis*, *E. alletteratus*, and *S. sarda*. These variations in nitrogen isotopic composition between different size classes may reflect changes in the consumption of prey from higher trophic levels as individuals grow or differences in foraging areas that are influenced by the origin of nutrients at the base of the food web (O'reilly et al. 2002). In the case of *T. atlanticus*, significant correlations exist between the preference of juvenile individuals for crustacean larvae and small fish, while adults tend to consume larger prey (Manooch and Mason 1983; Headley et al. 2009).

Dietary changes associated with the consumption of prey from higher trophic levels have also been reported for *K. pelamis* (Ankenbrandt 1985; Coletto et al. 2021), with a preferential increase in lanternfish consumption as the individual grows and a decrease in krill consumption off the southeast coast of Brazil. However, in another study in the same region, no significant changes were observed in the diet of *K. pelamis* with increasing size (Vilela 1990). Finally, positive correlations between standard length and $\delta^{15}\text{N}$ values for *S. sarda* and *E. alletteratus* in the current study may also be related to the consumption of prey from higher trophic levels by larger individuals, as has been documented for other regions (Campo et al. 2006; Falautano et al. 2007; Bahou et al. 2007; Genç et al. 2019), with a preferential shift towards the consumption of mesopelagic species in larger individuals. These changes may be related to the limitations of juvenile tunas in vertical migration and access to larger prey.

Changes in carbon isotopic composition with size may be associated with the use of different foraging areas or even the consumption of prey with different isotopic compositions (Nunn et al. 2012). In the present study, positive correlations were found between size and $\delta^{13}\text{C}$ for *T. albacares* and *C. hippurus*. The higher $\delta^{13}\text{C}$ values observed for *T. albacares* adults compared to juveniles may indicate an increased consumption of epipelagic fish, which have less depleted $\delta^{13}\text{C}$ values compared to cephalopods and crustaceans (Estrada et al. 2003; Li et al. 2016). This behavior has been previously reported for *T. albacares* in Ecuador and Hawaii, where individuals of larger classes fed preferentially on teleosts, whereas juveniles consumed crustaceans and cephalopods (Graham et al. 2007; Varela et al. 2017).

Considering that an increase in the consumption of epipelagic fish in the adult stage has not been reported for *C. hippurus*, but rather an increase in cephalopod consumption in its diet in the Baja California Peninsula region (Torres-Rojas et al. 2014),

the differences observed here for *C. hippurus* may be associated with the ability of adult individuals to consume a wide variety of prey and in foraging regions with different isoscape baselines, as was observed in the Central Pacific region (Briones-Hernández et al. 2023).

Author Contributions

B.N. Machado: conceptualization, methodology, formal analysis, validation, investigation, data curation, writing – original draft, visualization. **G. Montanini:** formal analysis, validation. **A.F. Azevedo:** resources, writing – review and editing, funding acquisition. **J. Lailson-Brito:** resources, writing – review and editing, funding acquisition. **T.L. Bisi:** conceptualization, methodology, resources, validation, writing – review and editing, supervision, funding acquisition, project administration.

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Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available on request.

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